

RoHS / REACH / Halogen-Free

Environmental compliance for PCB materials and assemblies

Substance restrictions that govern your BOM and material choice

Audience: Engineers shipping products globally

RoHS • REACH • WEEE • SVHC • Halogen-free • Compliance docs

Three Pillars of Substance Compliance

RoHS

Restriction of Hazardous Substances
10 banned substances
(Lead, Mercury, etc.)

REACH

Registration, Evaluation, Authorisation
of Chemicals
EU SVHC tracking

WEEE

Waste Electrical + Electronic
Equipment
EU producer responsibility

ELV

End-of-Life Vehicles
Automotive substance restrictions

Conflict minerals

Tin, Tantalum, Tungsten, Gold
Supply chain audit

Halogen-free

Bromine + chlorine in resin
For enclosed-space products

RoHS Banned Substances



Substance	Max allowable	Common source
Lead (Pb)	1000 ppm (0.1%)	Solder, brass, ceramics
Mercury (Hg)	1000 ppm	Fluorescent lamps, switches
Cadmium (Cd)	100 ppm	Yellow pigments, plating
Hexavalent chromium (Cr ⁶⁺)	1000 ppm	Anti-corrosion finishes
PBB (polybrominated biphenyls)	1000 ppm	Flame retardants
PBDE (polybrominated diphenyl ethers)	1000 ppm	Flame retardants
DEHP	1000 ppm	Plasticizer (cable jackets)
BBP	1000 ppm	Plasticizer
DBP	1000 ppm	Plasticizer
DIBP	1000 ppm	Plasticizer

RoHS Regions

- ▶ EU RoHS — mandatory for sale in EU (Directive 2011/65/EU, RoHS 3)
- ▶ Indian RoHS — under E-Waste Management Rules + BIS
- ▶ China RoHS — 'Management Methods for the Restriction...' SJ/T 11364
- ▶ USA: state-level RoHS in California (Prop 65, etc.); no federal mandate
- ▶ Japan: JIS C 0950 (J-RoHS); voluntary but practically mandatory
- ▶ Korea: K-RoHS, mandatory
- ▶ Most products shipped to multiple regions = comply with EU RoHS as baseline

REACH + SVHC

- ▶ REACH = Registration, Evaluation, Authorization, Restriction of Chemicals (EU regulation)
- ▶ SVHC = Substance of Very High Concern — current list ~240+ substances
- ▶ SVHC list updated every 6 months by ECHA
- ▶ If ANY SVHC > 0.1% in a homogeneous part of your product → reporting obligation
- ▶ Notify EU customers if SVHC content > 0.1%
- ▶ Add to SCIP database (EU portal for SVHC-containing articles)
- ▶ Common in PCBs: lead (in older HASL), some phthalates in cable jackets
- ▶ Component manufacturer should provide SVHC certificates

WEEE

- ▶ Waste Electrical and Electronic Equipment Directive (2012/19/EU)
- ▶ Producer responsibility for end-of-life electronics
- ▶ Mandatory: take-back schemes for electronic products
- ▶ WEEE symbol (crossed-out wheelie bin) on product or packaging
- ▶ Fee paid per kg of equipment placed on EU market
- ▶ Indian equivalent: E-Waste (Management) Rules 2022
- ▶ Recycling target: 65% of placed-on-market weight
- ▶ Cost: 0.5-2% of product cost in WEEE compliance fees

Conflict Minerals

- ▶ Four 'conflict minerals': Tin (Sn), Tantalum (Ta), Tungsten (W), Gold (Au) — '3TG'
- ▶ US Dodd-Frank Act: public companies disclose origin
- ▶ EU Regulation 2017/821: importer due diligence required
- ▶ Trace origin: DRC + adjoining countries are 'conflict-affected'
- ▶ Audit: ask component vendors for CMRT (Conflict Minerals Reporting Template)
- ▶ Free of conflict minerals = sourced from certified smelters
- ▶ Indian context: not yet regulated but global supply chain requires compliance

Halogen-Free PCBs

- ▶ Standard FR-4: brominated flame retardant + chlorine in resin (combined < 1500 ppm)
- ▶ Halogen-free FR-4: < 900 ppm Br + < 900 ppm Cl combined
- ▶ Required for: products with strict EU environmental specs (Apple, Dell, IKEA)
- ▶ Used in: enclosed-space products (medical, automotive interior, mil-spec)
- ▶ Cost: 10-30% premium over standard FR-4
- ▶ Slightly different mechanical properties — verify with fab
- ▶ Specify in fab quote: 'Halogen-free FR-4, IEC 61249-2-21'
- ▶ Not strictly required for RoHS — but a common 'green' upgrade

Compliance Documentation

- ▶ RoHS Declaration of Conformity (DoC): self-declared, kept on file 10 years
- ▶ Component data sheets must state RoHS / REACH status
- ▶ Material Declaration Sheet — comprehensive component listing
- ▶ IPC-1752 standard format for material declarations
- ▶ Test reports: XRF (X-Ray Fluorescence) testing for spot-checks
- ▶ Full ICP-MS / ICP-OES for accurate substance quantification
- ▶ Audit trail: keep all certificates from suppliers
- ▶ Cost: ₹5-20k for spot test; ₹50-100k for full chemical analysis

Common Compliance Issues

- ▶ Lead-bearing HASL on RoHS-required product → fail at customs
- ▶ Lead-bearing solder used by CM without confirmation → product non-compliant
- ▶ SVHC update changes status of a part → need new disclosure
- ▶ Counterfeit components — claim RoHS but actually contain lead
- ▶ Component EOL replaced with non-RoHS alternative → silent non-compliance
- ▶ WEEE mark missing on product → fine + recall
- ▶ Mixed RoHS / non-RoHS production runs → can't easily separate
- ▶ Conflict mineral disclosure not maintained → liability under Dodd-Frank

Best Practices

- ▶ Spec lead-free SOLDER + lead-free HASL or ENIG/OSP from day 1
- ▶ Use High-Tg FR-4 (170 °C+) for lead-free reflow
- ▶ Component RoHS certs in your BOM database — track per MPN
- ▶ Halogen-free if shipping to certain corporate customers
- ▶ Spot XRF test on incoming materials + finished boards
- ▶ Update SVHC list quarterly — re-validate component compliance
- ▶ Plan WEEE marking + scheme registration if shipping to EU
- ▶ Conflict mineral CMRT update annually

Cost Impact

- ▶ Lead-free solder: ~30% more expensive than Sn-Pb
- ▶ Lead-free HASL: same cost as leaded HASL today
- ▶ High-Tg FR-4 for lead-free: 20-50% more than standard FR-4
- ▶ Halogen-free PCB: 10-30% premium
- ▶ Compliance testing: ₹5-50k per design depending on scope
- ▶ WEEE scheme fees: 0.5-2% of EU sales revenue
- ▶ Documentation overhead: 1-2 engineering days per product
- ▶ Compared to fines + recalls: compliance is dirt-cheap insurance

Key Takeaways

Key takeaways from this lecture

RoHS = baseline

Lead-free + 9 other restrictions.

REACH SVHC quarterly

List grows; re-audit.

WEEE for EU

Marking + take-back scheme.

Halogen-free for premium

Strict corporate customers.

Document everything

Certs from each component.

Plan early

Retrofitting is expensive.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- ▶ EU Directive 2011/65/EU (RoHS 3) full text
- ▶ ECHA SVHC list (echa.europa.eu)
- ▶ IPC-1752 Material Declaration standard
- ▶ BIS India E-Waste rules + RoHS section
- ▶ China RoHS SJ/T 11364
- ▶ Conflict Minerals Reporting Template (CMRT) — RMI
- ▶ Dodd-Frank Section 1502 (USA)
- ▶ Indian standards on RoHS — BIS website

Thank you • Build something this week.